Lab -1

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EXPERIMENT 1

Objective / Aim

1. To check the working of NOT gate using HEX Inverter

Components Used

- A LED
- A HEX inverter (for NOT gate)
- Battery (1.5v)
- wires(red,black and green)
- bread board

Hypothesis

NOT gate works on the principle that it gives the opposite output of what we put as Input.
Example. If we give 1 as an input NOT gate makes it 0.

When we do not use a hex inverter to run the LED it lights up as it should but when we use a hex inverter it changes the input 1 to input 0. So, the LED does not lights up.

Here, input 1 means, supply of 1.5v of electricity and 0 input means no supply of electricity.
We connect negative end of battery to hex inverter ground to make sure that both have same potential and the circuit is completed.

Procedure

- Step 1 – Connect the hex inverter to the centre of breadboard like it is connected to both upper set of vertical strip as well lower set of vertical strip.
- Step 2 – We connect one end of red wire with the positive end of battery and the other end is attached to the power input of hex inverter.

- Step 3 – Now, we connect the black wire’s one end with the negative end of battery and the other end to the ground hex inverter.
- Step 4 – We connect one end of a wire in input 1 of hex inverter (or one of the holes in the vertical strip) and the other end in some other vertical strip which is not a part of hex inverter’s vertical strip.
- Step 5 – We connect another wire to output 1 and its other end adjacent to our previous wire but not in the same vertical strip.(either on left or right)
- Step 6 – Then we connect the LED’s anode to the same vertical strip where the wire with input 1 ended.
- Step 7 – Then we connect the LED’s cathode to the same vertical strip where the wire with output 1 ended
- Step 8 - We repeat the above procedure but without using hex invertrer. We just directly put LED’s cathode in the same verical strip as the black(negative end of battery’s) wire and the anode in the same vertical strip as the red (positive end of battery’s) wire.
- Step – 9 Now, we switch on the battery.

Observations

With hex inverter

Input	Output
Electricity given (1)	LED does not glows (0)
No electricity given (0)	LED glows (1)

Without hex inverter

Input	Output
Electricity given (1)	LED glows (1)
No electricity given (0)	LED does not glows (0)

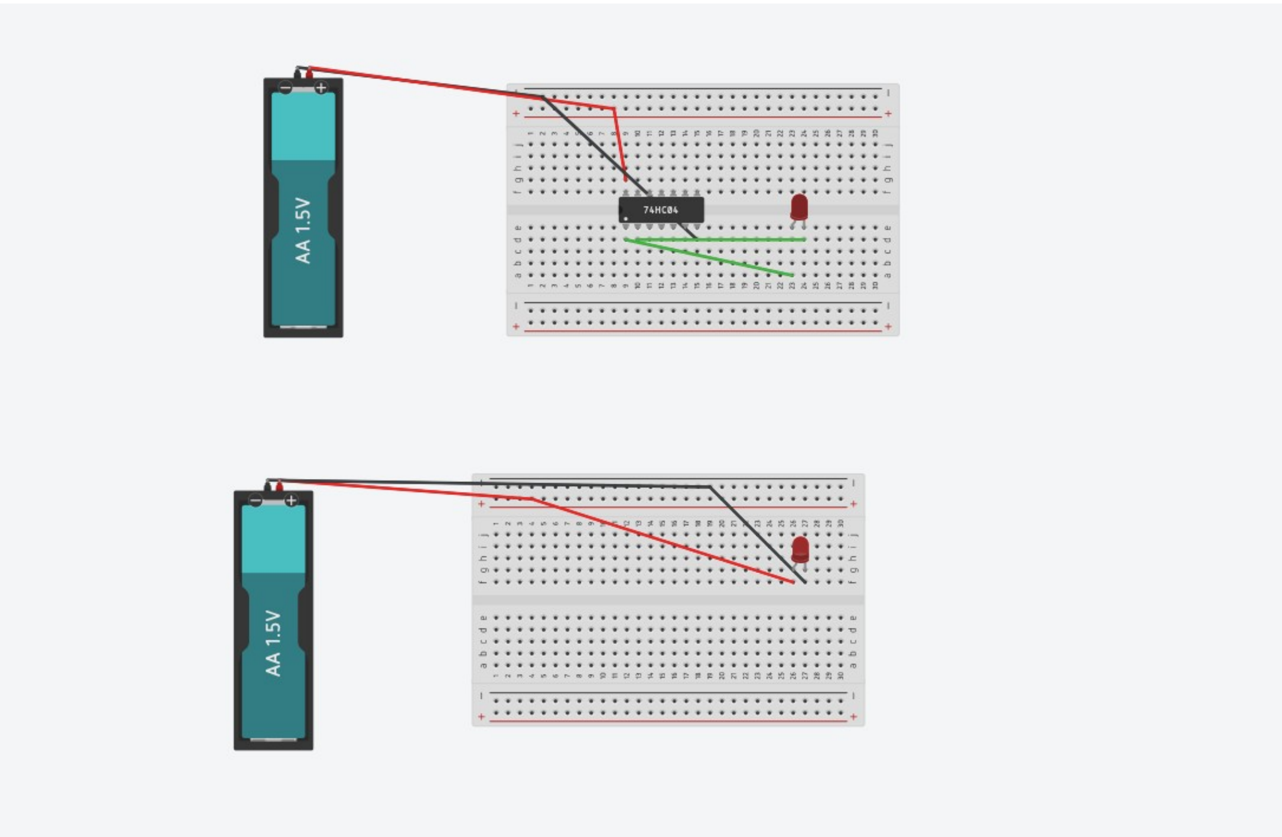
When we connect circuit without hex inverter the LED glows when battery is connected and switches off when we disconnect it.

But in the hex inverter circuit the LED does not glows us when provided with electricity as the NOT gate in the hex inverter invertes the input from 1 to 0. Similarly when battery is off then the LED glows because hex inverter converts input 0 to 1. (provided that circuit is not broken and complete)

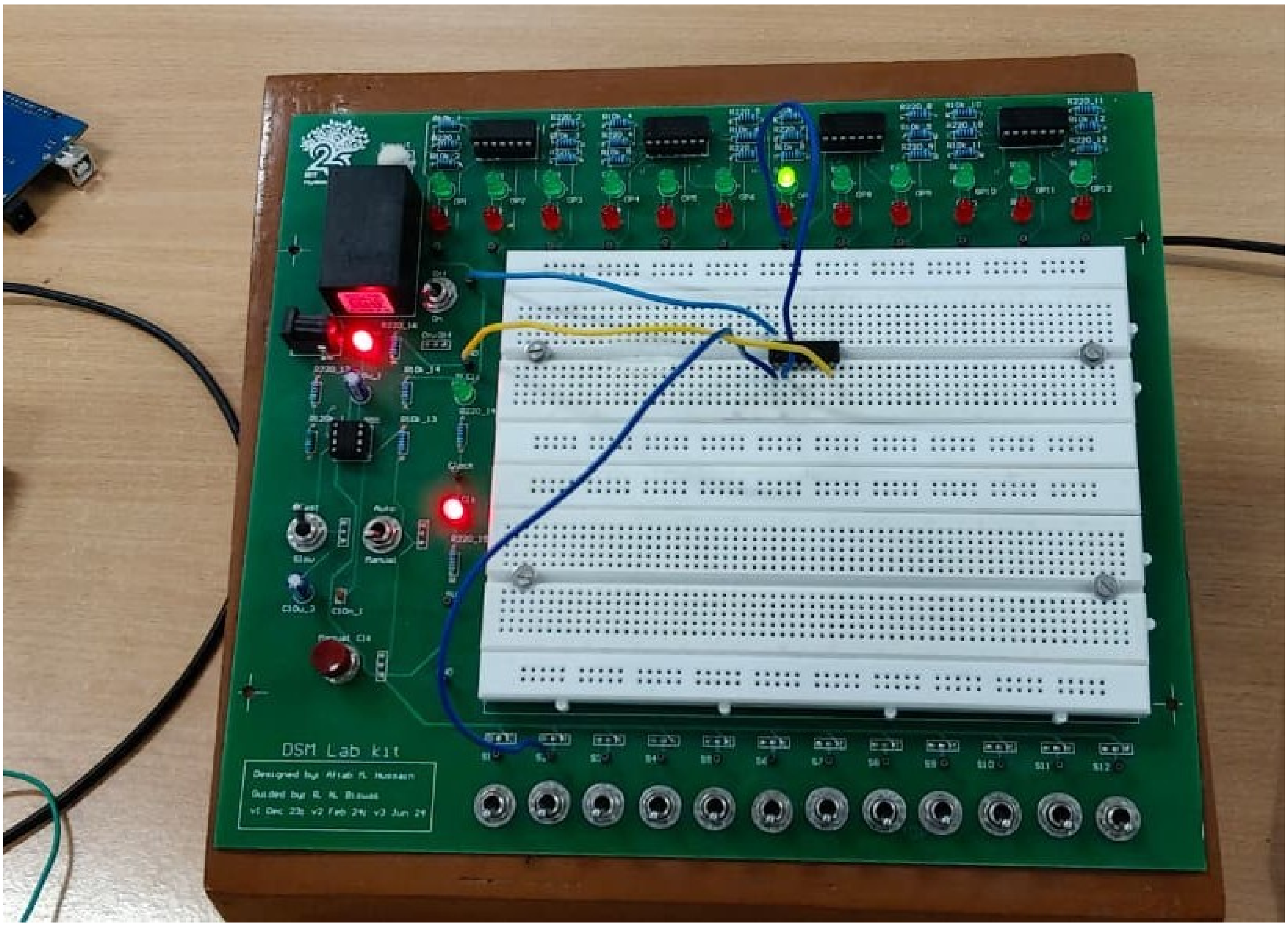
Simulation Link(TinkerCAD)

https://www.tinkercad.com/things/1q3ZN4jho0D-dsm-lab-1-experiment-2/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard%2Fdesigns%2Fcircuits&sharecode=UYq59__pR7aFqQ8YsPQGkIvEgaGZVlv712TwmcmrFrY

TinkerCAD Screenshot



Lab Photograph



Conclusion

The hex inverter acts like a NOT gate which changes binary input 0 to 1 and similarly input 1 to 0.
This causes the LED not to glow even when connected to electricity as the hex inverter inverts the input.